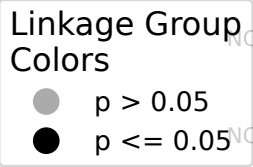


diamond: 9274 orthologs on 3932 Rphi scaffolds and 9 Pvul scaffolds.



Rphi vs P Rphi scaffolds

Rphi vs Pvul FET, whole chroms.

	Rphi scaf	Pvul scaf	p (FET)	Count
0	NW_026865571.1	NC_065883.2	3.61e-14	19
1	NW_026861563.1	NC_065883.2	4.89e-11	18
2	NW_026861019.1	NC_065883.2	4.89e-11	18
3	NW_026852258.1	NC_065882.2	3.42e-12	18
4	NW_026860903.1	NC_065880.2	2.26e-09	17
5	NW_026851633.1	NC_065881.2	1.32e-08	17
6	NW_026865484.1	NC_065883.2	5.64e-07	15
7	NW_026861084.1	NC_065881.2	4.97e-07	14
8	NW_026853802.1	NC_065880.2	6.62e-07	14
9	NW_026852008.1	NC_065880.2	6.62e-07	14
10	NW_026865967.1	NC_065882.2	9.85e-08	13
11	NW_026862783.1	NC_065880.2	3.58e-07	13
12	NW_026862663.1	NC_065880.2	4.17e-03	13
13	NW_026854960.1	NC_065882.2	1.21e-06	13
14	NW_026855294.1	NC_065881.2	1.60e-02	12
15	NW_026853170.1	NC_065880.2	2.85e-05	12
16	NW_026852705.1	NC_065880.2	2.85e-05	12
17	NW_026863896.1	NC_065881.2	1.41e-05	11
18	NW_026861130.1	NC_065881.2	8.39e-04	11
19	NW_026859790.1	NC_065883.2	1.52e-05	11
20	NW_026858503.1	NC_065885.2	2.23e-05	11
21	NW_026857707.1	NC_065881.2	1.48e-04	11
22	NW_026852000.1	NC_065879.2	1.92e-06	11
23	NW_026867184.1	NC_065885.2	3.49e-05	10
24	NW_026863541.1	NC_065883.2	6.64e-04	10
25	NW_026863406.1	NC_065885.2	3.48e-06	10
26	NW_026862474.1	NC_065880.2	1.19e-03	10
27	NW_026861574.1	NC_065880.2	6.24e-03	10
28	NW_026859998.1	NC_065882.2	4.49e-04	10
29	NW_026859959.1	NC_065887.2	4.92e-08	10
30	NW_026859933.1	NC_065882.2	2.38e-03	10
31	NW_026859561.1	NC_065884.2	9.26e-06	10
32	NW_026859418.1	NC_065882.2	4.62e-05	10
33	NW_026859085.1	NC_065880.2	1.19e-03	10
34	NW_026855282.1	NC_065882.2	4.49e-04	10
35	NW_026852468.1	NC_065883.2	1.23e-04	10
36	NW_026864590.1	NC_065880.2	8.75e-04	9
37	NW_026864352.1	NC_065882.2	1.54e-02	9
38	NW_026864086.1	NC_065879.2	1.42e-04	9
39	NW_026862956.1	NC_065883.2	4.90e-03	9
40	NW_026862689.1	NC_065883.2	4.90e-03	9
41	NW_026861864.1	NC_065881.2	3.06e-02	9
42	NW_026861840.1	NC_065884.2	3.77e-03	9
43	NW_026861602.1	NC_065882.2	3.59e-04	9
44	NW_026859757.1	NC_065882.2	3.59e-04	9
45	NW_026859313.1	NC_065880.2	8.75e-04	9
46	NW_026859162.1	NC_065881.2	7.27e-04	9
47	NW_026857461.1	NC_065881.2	7.27e-04	9
48	NW_026857221.1	NC_065884.2	3.77e-03	9
49	NW_026855427.1	NC_065887.2	7.60e-07	9
50	NW_026853973.1	NC_065883.2	1.10e-04	9
51	NW_026853925.1	NC_065884.2	8.44e-05	9
52	NW_026852998.1	NC_065880.2	3.66e-02	9
53	NW_026852612.1	NC_065884.2	8.44e-05	9
54	NW_026851646.1	NC_065881.2	3.06e-02	9
55	NW_026867092.1	NC_065880.2	6.14e-03	8
56	NW_026866929.1	NC_065882.2	9.81e-02	8
57	NW_026866837.1	NC_065883.2	3.56e-02	8
58	NW_026866672.1	NC_065884.2	7.69e-04	8
59	NW_026866451.1	NC_065882.2	2.22e-02	8
60	NW_026866212.1	NC_065880.2	6.14e-03	8
61	NW_026866098.1	NC_065880.2	6.14e-03	8
62	NW_026866080.1	NC_065883.2	9.77e-04	8
63	NW_026865813.1	NC_065882.2	2.78e-03	8

RphRphRphRphRph FET, chrom. pieces

	Rphi scaf	Pvul scaf	p (FET)	Count
0	NW_026865571.1 : 0-end	NC_065883.2 : 0-end	3.61e-14	19
1	NW_026861563.1 : 0-end	NC_065883.2 : 0-end	4.89e-11	18
2	NW_026861019.1 : 0-end	NC_065883.2 : 0-end	4.89e-11	18
3	NW_026852528.1 : 0-end	NC_065882.2 : 0-end	3.42e-12	18
4	NW_026860903.1 : 0-end	NC_065880.2 : 0-end	2.26e-09	17
5	NW_026851633.1 : 0-end	NC_065881.2 : 0-end	1.32e-08	17
6	NW_026865484.1 : 0-end	NC_065883.2 : 0-end	5.64e-07	15
7	NW_026861084.1 : 0-end	NC_065881.2 : 0-end	4.97e-07	14
8	NW_026853802.1 : 0-end	NC_065880.2 : 0-end	6.62e-07	14
9	NW_026852008.1 : 0-end	NC_065880.2 : 0-end	6.62e-07	14
10	NW_026865967.1 : 0-end	NC_065882.2 : 0-end	9.85e-08	13
11	NW_026862783.1 : 0-end	NC_065880.2 : 0-end	3.58e-07	13
12	NW_026862663.1 : 0-end	NC_065880.2 : 0-end	4.17e-03	13
13	NW_026854960.1 : 0-end	NC_065882.2 : 0-end	1.21e-06	13
14	NW_026855294.1 : 0-end	NC_065881.2 : 0-end	1.60e-02	12
15	NW_026853170.1 : 0-end	NC_065880.2 : 0-end	2.85e-05	12
16	NW_026852705.1 : 0-end	NC_065880.2 : 0-end	2.85e-05	12
17	NW_026863896.1 : 0-end	NC_065881.2 : 0-end	1.41e-05	11
18	NW_026861130.1 : 0-end	NC_065881.2 : 0-end	8.39e-04	11
19	NW_026859790.1 : 0-end	NC_065883.2 : 0-end	1.52e-05	11
20	NW_026858503.1 : 0-end	NC_065885.2 : 0-end	2.23e-05	11
21	NW_026857770.1 : 0-end	NC_065881.2 : 0-end	1.48e-04	11
22	NW_026852000.1 : 0-end	NC_065879.2 : 0-end	1.92e-06	11
23	NW_026867184.1 : 0-end	NC_065885.2 : 0-end	3.49e-05	10
24	NW_026863541.1 : 0-end	NC_065883.2 : 0-end	6.64e-04	10
25	NW_026863406.1 : 0-end	NC_065885.2 : 0-end	3.48e-06	10
26	NW_026862474.1 : 0-end	NC_065880.2 : 0-end	1.19e-03	10
27	NW_026861574.1 : 0-end	NC_065880.2 : 0-end	6.24e-03	10
28	NW_026859998.1 : 0-end	NC_065882.2 : 0-end	4.49e-04	10
29	NW_026859959.1 : 0-end	NC_065887.2 : 0-end	4.92e-08	10
30	NW_026859933.1 : 0-end	NC_065882.2 : 0-end	2.38e-03	10
31	NW_026859561.1 : 0-end	NC_065884.2 : 0-end	9.26e-06	10
32	NW_026859418.1 : 0-end	NC_065882.2 : 0-end	4.62e-05	10
33	NW_026859085.1 : 0-end	NC_065880.2 : 0-end	1.19e-03	10
34	NW_026855828.1 : 0-end	NC_065882.2 : 0-end	4.49e-04	10
35	NW_026852468.1 : 0-end	NC_065883.2 : 0-end	1.23e-04	10
36	NW_026864590.1 : 0-end	NC_065880.2 : 0-end	8.75e-04	9
37	NW_026864352.1 : 0-end	NC_065882.2 : 0-end	1.54e-02	9
38	NW_026864086.1 : 0-end	NC_065879.2 : 0-end	1.42e-04	9
39	NW_026862956.1 : 0-end	NC_065883.2 : 0-end	4.90e-03	9
40	NW_026862689.1 : 0-end	NC_065883.2 : 0-end	4.90e-03	9
41	NW_026861864.1 : 0-end	NC_065881.2 : 0-end	3.06e-02	9
42	NW_026861840.1 : 0-end	NC_065884.2 : 0-end	3.77e-03	9
43	NW_026861602.1 : 0-end	NC_065882.2 : 0-end	3.59e-04	9
44	NW_026859757.1 : 0-end	NC_065882.2 : 0-end	3.59e-04	9
45	NW_026859313.1 : 0-end	NC_065880.2 : 0-end	8.75e-04	9
46	NW_026859162.1 : 0-end	NC_065881.2 : 0-end	7.27e-04	9
47	NW_026857461.1 : 0-end	NC_065881.2 : 0-end	7.27e-04	9
48	NW_026857221.1 : 0-end	NC_065884.2 : 0-end	3.77e-03	9
49	NW_026855427.1 : 0-end	NC_065887.2 : 0-end	7.60e-07	9
50	NW_026853973.1 : 0-end	NC_065883.2 : 0-end	1.10e-04	9
51	NW_026853925.1 : 0-end	NC_065884.2 : 0-end	8.44e-05	9
52	NW_026852998.1 : 0-end	NC_065880.2 : 0-end	3.66e-02	9
53	NW_026852612.1 : 0-end	NC_065884.2 : 0-end	8.44e-05	9
54	NW_026851646.1 : 0-end	NC_065881.2 : 0-end	3.06e-02	9
55	NW_026867092.1 : 0-end	NC_065880.2 : 0-end	6.14e-03	8
56	NW_026866929.1 : 0-end	NC_065882.2 : 0-end	9.81e-02	8
57	NW_026866837.1 : 0-end	NC_065883.2 : 0-end	3.56e-02	8
58	NW_026866672.1 : 0-end	NC_065884.2 : 0-end	7.69e-04	8
59	NW_026866451.1 : 0-end	NC_065882.2 : 0-end	2.22e-02	8
60	NW_026866212.1 : 0-end	NC_065880.2 : 0-end	6.14e-03	8
61	NW_026866098.1 : 0-end	NC_065880.2 : 0-end	6.14e-03	8
62	NW_026866080.1 : 0-end	NC_065883.2 : 0-end	9.77e-04	8
63	NW_026865813.1 : 0-end	NC_065882.2 : 0-end	2.78e-03	8

(a) depicts the chromosomes/scaffolds of Rphi (x-axis) plotted against the chromosomes/scaffolds of Pvul (y axis). Each dot in the plot represents an ortholog, specifically a reciprocal best diamond blastp match between two species. The unit of the x- and y-axes are the number of orthologous proteins between these two species: 9274 orthologs found between 3932 Rphi scaffolds and 9 Pvul scaffolds. If there are chromosome breaks, Fisher's exact test is used to calculate the significance of the interactions between the sub-chromosomeal pieces. Otherwise Fisher's exact test is calculated on whole chromosomes. The opacity of the dots depict the significance from Fisher's exact test. Dots that are a solid color are in cells with a FET p-value less than or equal to 0.05. Dots that are translucent are in cells with a FET p-value greater than 0.05. The Dx and Dy values depict places where there may be sudden breaks in synteny. See the supplementary information the following paper for more information on Dx and Dy: Simakov, Oleg, et al. Deeply conserved synteny resolves early events in vertebrate evolution. Nature Ecology & Evolution 4.6 (2020): 820-830. (b) depicts the same information as panel a, but it is plotted in the organisms chromosome basepair coordinates rather than gene index. This is useful for visualizing gene-poor regions of the chromosomes. (c) shows which gene groups are most prevalent in each chromosome pair. The FET p-value in this table corresponds to the whole-chromosome FET p-value , and is not a FET value of the correlation between the chromosome pair and the gene group. (d) shows chromosome-scale significance values. This table shows the same information as c, but does not show the gene group. (e) shows the FET p-values for each chromosome pair. (f) shows the FET p-values for each chromosome pair. (g) shows the FET p-values for each chromosome pair.

	NW_026860663.1	NC_055882.2	FET p-value	Dx	Dy
74	NW_026860663.1	NC_055882.2	1.0E-04	0	0
75	NW_026860663.1	NC_055882.2	2.78E-03	0	0

	NW_026860663.1	NC_055882.2	FET p-value	Dx	Dy
74	NW_026860663.1	NC_055882.2	1.0E-04	0	0
75	NW_026860663.1	NC_055882.2	2.78E-03	0	0

	NW_026860663.1	NC_055882.2	FET p-value	Dx	Dy
74	NW_026860663.1	NC_055882.2	1.0E-04	0	0
75	NW_026860663.1	NC_055882.2	2.78E-03	0	0

	NW_026860663.1	NC_055882.2	FET p-value	Dx	Dy
74	NW_026860663.1	NC_055882.2	1.0E-04	0	0
75	NW_026860663.1	NC_055882.2	2.78E-03	0	0

	NW_026860663.1	NC_055882.2	FET p-value	Dx	Dy
74	NW_026860663.1	NC_055882.2	1.0E-04	0	0
75	NW_026860663.1	NC_055882.2	2.78E-03	0	0

	NW_026860663.1	NC_055882.2	FET p-value	Dx	Dy
74	NW_026860663.1	NC_055882.2	1.0E-04	0	0
75	NW_026860663.1	NC_055882.2	2.78E-03	0	0